# EcoLoop

Circular material systems and zero-cost digital architecture.

Swami Sant Dass Public School, Jalandhar

# The Packaging Trap: Multi-Layer Plastics

Why standard municipal recyclers throw clean packaging straight into landfills.

## The Material Reality

- **Bonded Metal & Plastic:**
  Chips and snack packets laminate aluminum foil between plastic film. Separating those layers is chemically difficult and commercially unprofitable.
- **100% Landfill Destination:**
  Because municipal plants cannot process multi-layer packaging, almost all snack wrappers are burned or dumped into local landfills.
- **Clean Waste in Schools:**
  Canteens and classrooms generate sacks of clean snack wrappers daily with zero realistic collection logistics.

## The EcoLoop Approach

- **Skip Separation, Heat-Compress:**
  Instead of separating the foil, we partner to heat-press whole wrappers into dense boards that replace plywood.
- **Unified Circular Model:**
  Unifying scattered school green crafts and outsourced heavy materials into one clean system.
- **Digital Transparency:**
  Building an online catalog so buyers can see exact specifications and request real samples.

# The Physical Loop

Six circular material systems systematically diverted from school waste streams.

STUDENT-MADE

### Seed Paper

Handmade plantable paper embedded with native seeds, crafted directly inside school.

STUDENT-MADE

### Bioenzymes

Citrus fruit peels fermented into natural multi-purpose cleaning liquids for campus sanitation.

STUDENT-MADE

### Uniform Bags

Durable, reusable tote bags stitched from discarded school uniform fabrics.

STUDENT-MADE

### Eco-Bricks

PET bottles tightly packed with clean plastic wrappers for modular walls and simple structures.

STUDENT-MADE

### Bottle Planters

Upcycled beverage containers for school corridor nursery and vertical greenery installations.

NGO PARTNER

### MLP Sheets

Heavy-duty structural boards heat-pressed from snack packaging using industrial hydraulic machinery.

# how it started vs how it's going



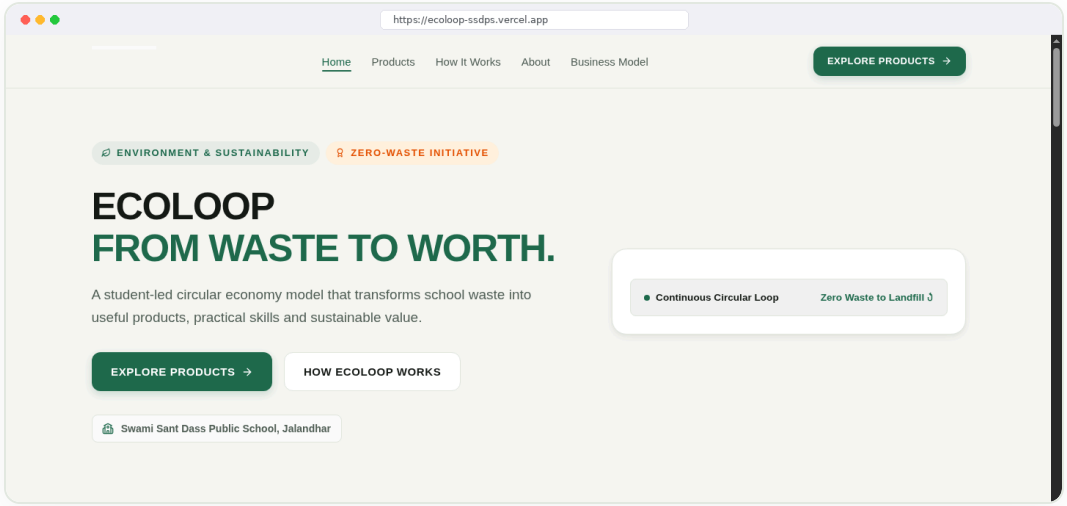

started as scattered classroom craft projects. turned into a unified circular brand with external partnerships and custom software.

# Why Build a Digital Platform?

Transforming school crafts into verifiable, specification-backed utility products.

## Platform Objectives

- **Engineering Credibility:**
  Publishing actual dimensions, material density, thermal ratings, and production lead times so buyers take recycled goods seriously.
- **Direct Inquiry Routing:**
  Giving institutions and partners a friction-free way to query available stock and request physical samples.
- **Connecting School to Outside World:**
  Taking activities happening inside classrooms and presenting them professionally to external organizations.



# The Zero-Cost Backend: Built on Google Sheets

Handling live incoming inquiries without paid servers or corporate credit cards.

STEP 01

### React Form

User fills in their name, product choice, and notes. The browser validates all inputs locally.

→

STEP 02

### Apps Script Webhook

A free Google Apps Script webhook receives the data payload securely in the cloud.

→

STEP 03

### Sheets Database

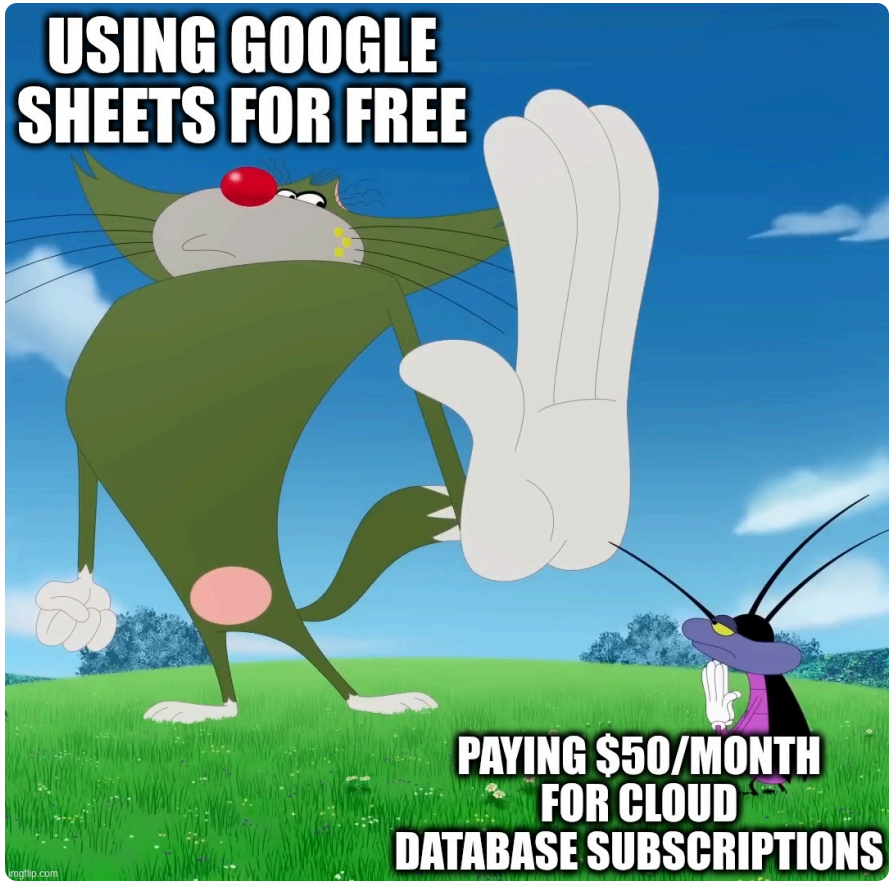
Appends the inquiry row into a live Google Sheet and immediately triggers an email alert.

**Monthly Infrastructure Bill**

No credit cards, zero server maintenance, and reliable for real order leads.

**$0.00 / month**

# why our database bill is literally $0



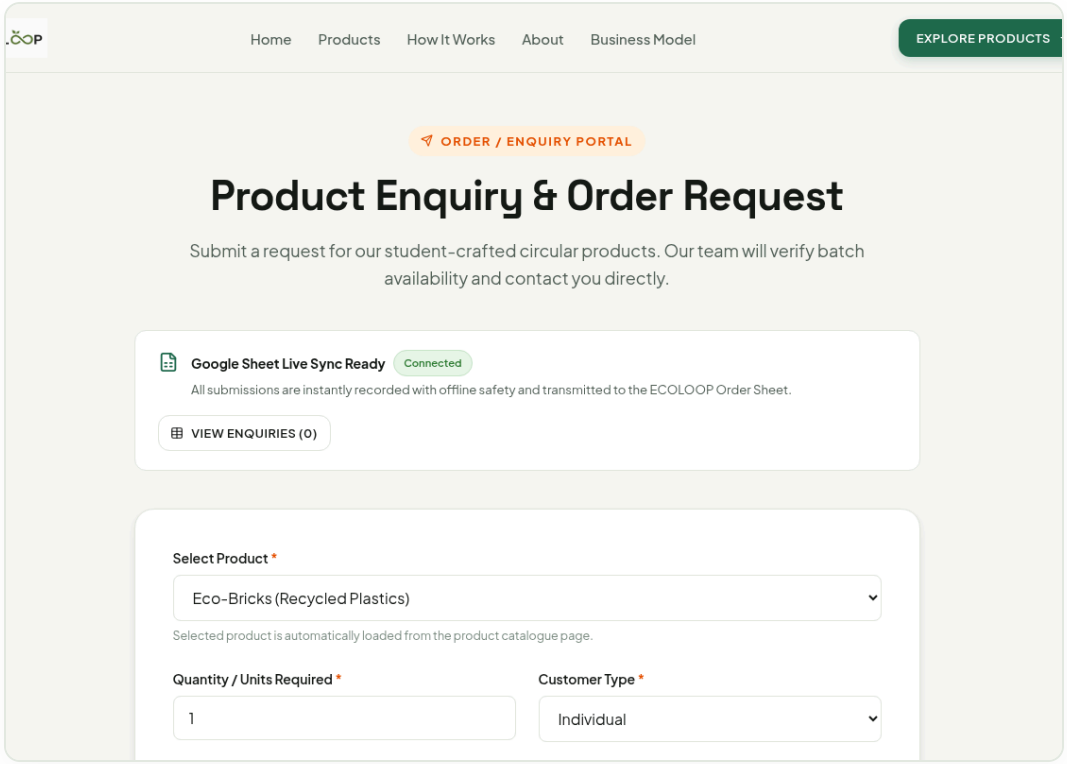

why spend monthly money on complex database servers when a free google sheet captures all your incoming inquiries?

# The Inquiry Engine: An Honest Review

Built in a single night sprint: functional, basic, and with plenty of room to improve.

## The Reality of Our MVP

- **Built in One Night:**
  Put together in a single late-night sprint to give the exhibition judges a working demo to test.
- **Basic but Working:**
  Automatically pre-populates product details and logs records straight into our backend without paying for services.
- **Room for Improvement:**
  Needs automated inventory deduction, better error states, and order tracking. It's an MVP, not final software.



# How I Built It Without Knowing React

Moving from typing boilerplate code to designing systems with AI agents.

## The Traditional Hurdle

- Getting stuck on missing brackets, package version mismatches, and unfamiliar framework setup.
- Spending entire evenings digging through Stack Overflow threads instead of building real features.
- Giving up on software ideas simply because you haven't spent 6 months memorizing syntax.

## Developer as Architect

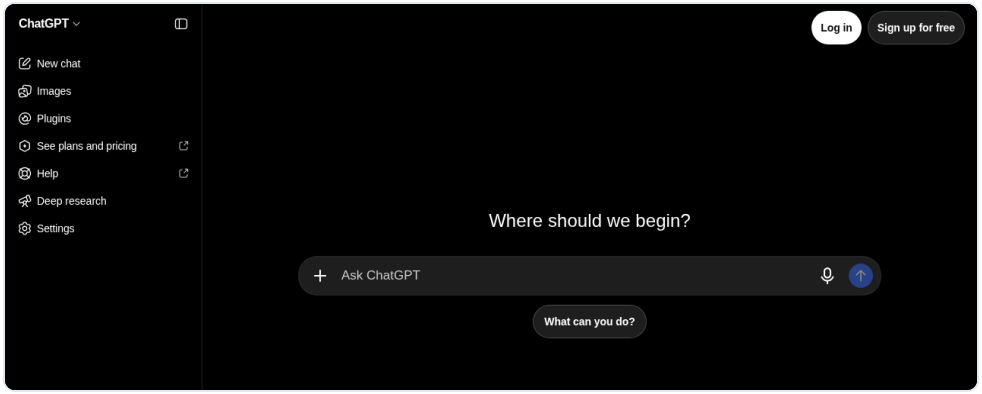
- You specify the data models, visual style, and constraints in plain language.
- An autonomous agent creates components, connects routes, and modifies multiple files.
- Your primary job becomes quality control, testing user flows, and refining the architecture.

# Chatbot vs Agent Workspace

The difference between copy-pasting snippets and letting an agent build inside your files.

## ChatGPT: Web Chatbot

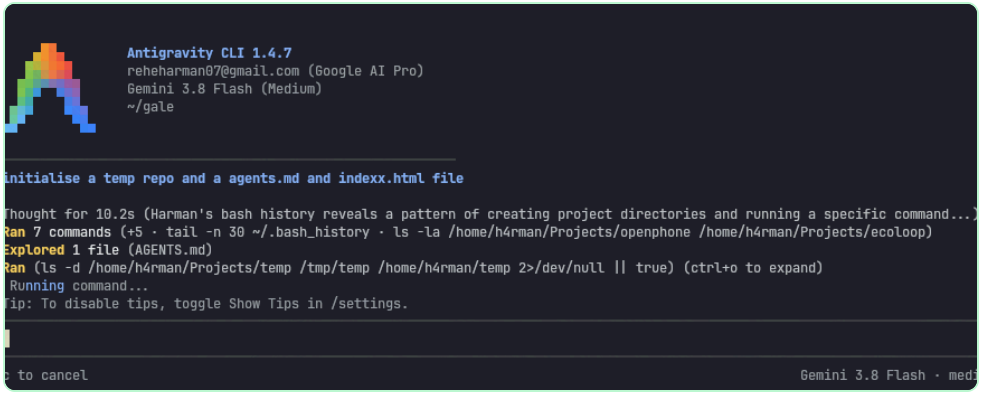
ISOLATED




- **Copy-Paste Loop:** Generates isolated code snippets; you must manually paste and fix them.
- **Zero File Access:** Cannot see your directory tree, configs, or local dependencies.
- **Blind Execution:** Cannot run builds, view terminal errors, or verify live changes.

## Antigravity: Agent Workspace

AUTONOMOUS




- **Direct Repository Edits:** Reads folder context and writes multi-file modifications directly.
- **Autonomous Terminal:** Executes shell commands, dev servers, and build scripts.
- **Self-Repair Loop:** Catches compiler and runtime errors and fixes them automatically.

# How Working with an Agent Actually Looks

A practical 4-step loop of specifying, editing, testing, and self-repairing.

STAGE 01

### Write the Spec

Describe what you want in plain language: colors, component layout, and behavior rules.

STAGE 02

### Agent Edits Files

The agent inspects your repository, writes components, and updates project files directly.

STAGE 03

### You Test Locally

You run the app in your browser, click the buttons, and see what works or what broke.

STAGE 04

### Agent Fixes Bugs

You paste the error back to the agent; it reads the log and repairs the broken code.

# Replicating This Workflow: OpenCode

An open-source terminal agent that gives you the same autonomous capabilities.

## Why OpenCode for Students?

- **100% Free & Open Source:**
  Created by Anomaly (GitHub: `anomalyco/opencode`). Zero subscriptions or vendor lock-in.
- **Lives in Your Terminal:**
  Reads your project files, edits code, and runs local commands just like specialized agent harnesses.
- **Dual Execution Modes:**
  Plan Mode reviews architecture safely; Build Mode executes real code changes.

```
// Quick Installation in 60 Seconds

$ npm install -g @opencode/cli
$ cd ./my-project
$ opencode

Runs on macOS, Linux, and Windows WSL.
```

# Free AI Models: Zero Cost for Students

Practical options you can use tonight without paying for subscriptions.

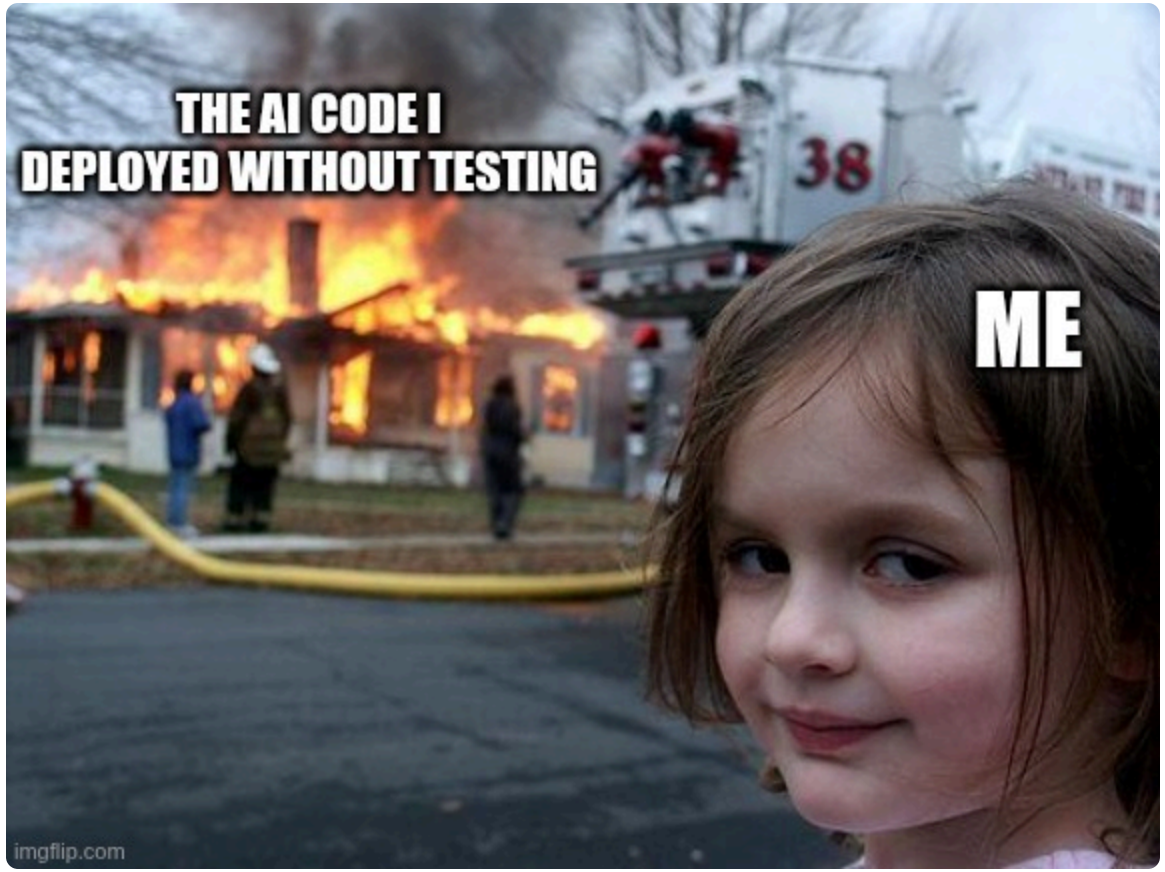
## Free Cloud APIs

- **Google Gemini API:**
  Sign in to `aistudio.google.com` for a free key. Generous free limits for fast coding models.
- **Groq Cloud:**
  Ultra-fast free inference on open models like Llama 3.3 via `console.groq.com`.
- **GitHub Models:**
  Free student access to preview models inside VS Code and GitHub Codespaces.

## 100% Offline with Ollama

- **Private & Local:**
  Download `ollama.com`. Runs entirely on your laptop hardware without needing internet.
- **Recommended Model:**
  `qwen2.5-coder:7b` delivers strong coding reasoning on standard student laptops.
- **Zero Latency:**
  Fast responses directly on your machine with complete data privacy.

# deployed without testing




prompting is only 20% of the work. clicking every button to see what actually broke is the other 80%.

# Thank You.

Scan the code to access the complete open-source repository, live demo, and student setup guide.



`github.com/Harman209/ecoloop`

**Swami Sant Dass Public School, Jalandhar**